

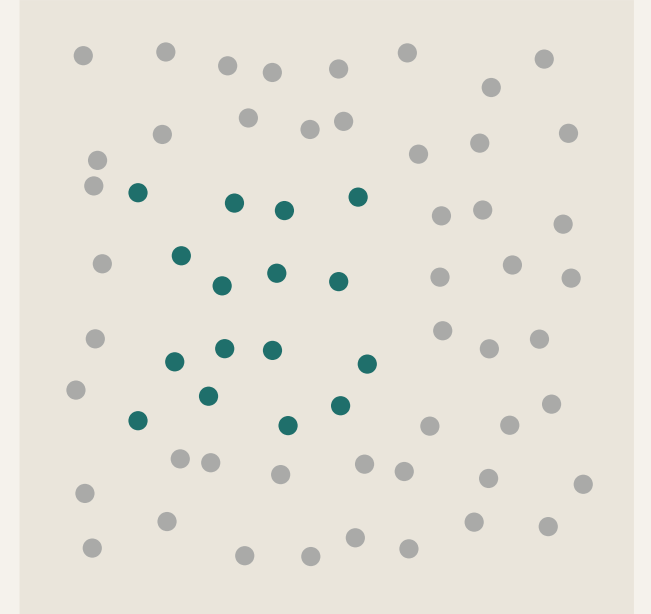
Fashion search: what the literature says

Recent papers mapped to our agenda

Compiled semantic predicates
RSA

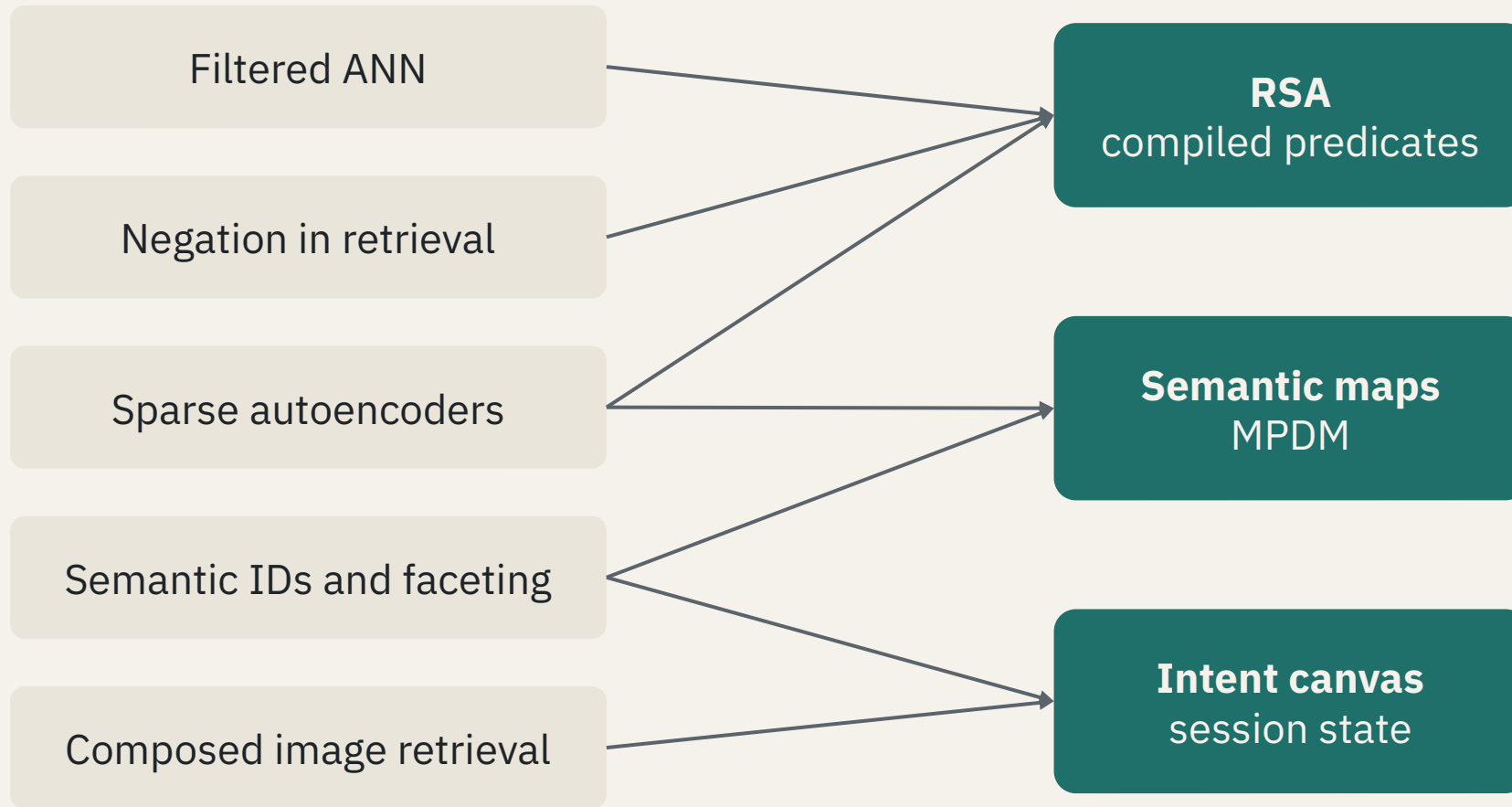
Semantic maps
MPDM

Intent canvas
Session state



formal – casual × minimal – ornate

Five research fronts feed our three ideas

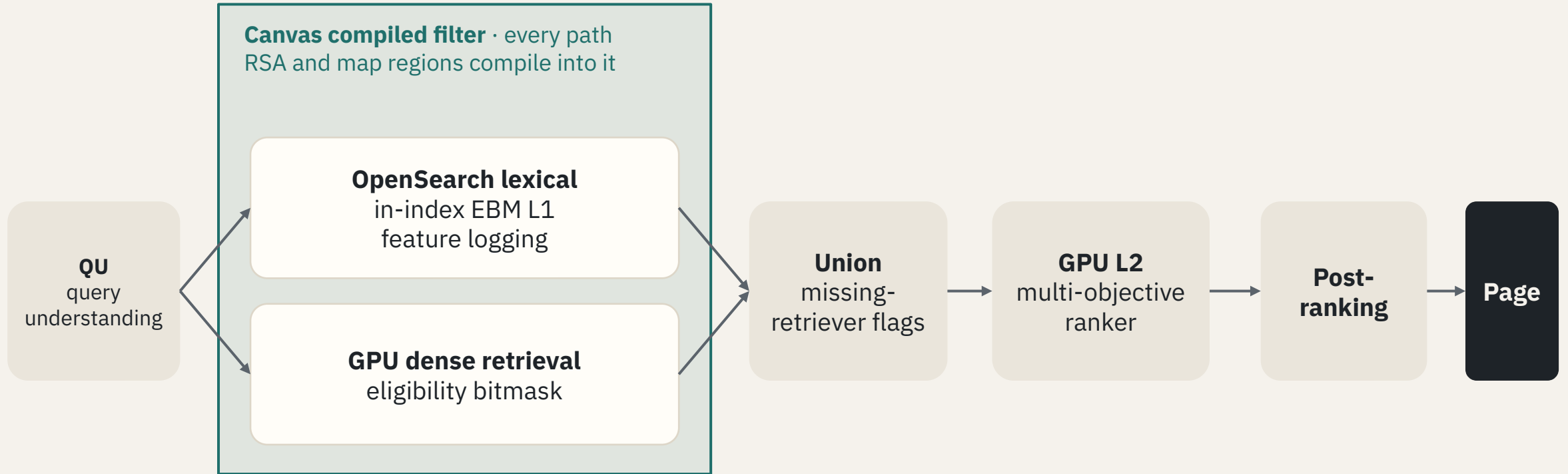


RSA draws on three fronts: filtered ANN, negation and SAEs.

Maps borrow from SAEs and semantic IDs.

Canvas borrows from semantic IDs and composed retrieval.

Where each idea plugs into our search stack



The canvas compiles once per turn and constrains **every** retriever, lexical and dense alike. Missing-retriever flags let L2 score items that only one path found.

RSA compiles a concept into a ~216-byte program

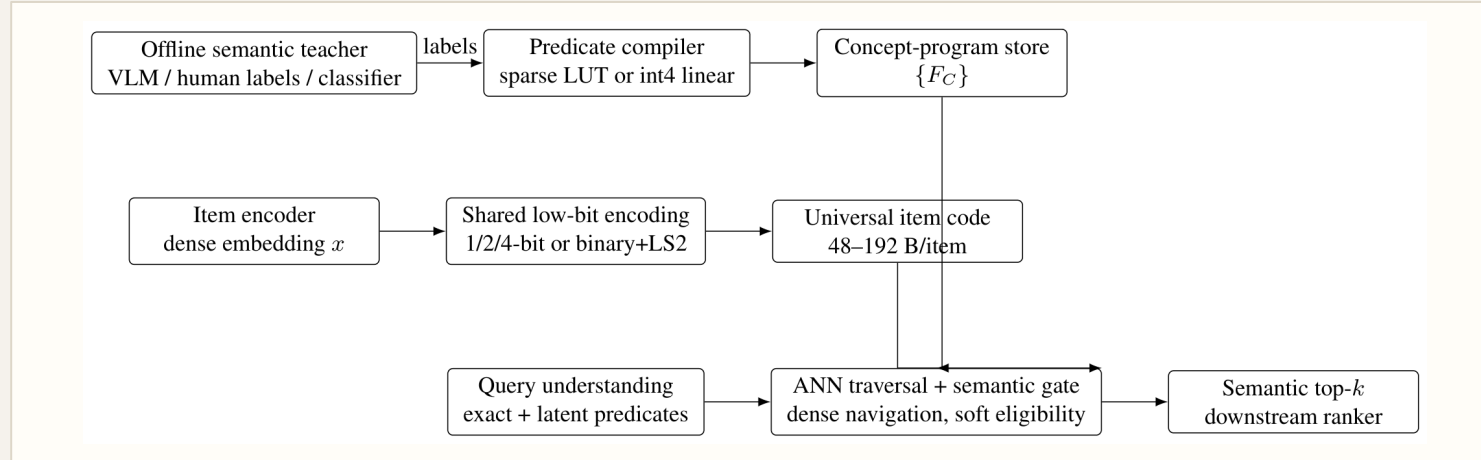


Figure 1 from Al-Shater (2026), local PDF, p. 2. License: author's own.

$$p_C(x) = \sigma(a'_C F_C(x) + c_C) \quad M(N, K) = N B_{\text{item}} + K B_{\text{program}}$$

$$S_Q(x) = \sum_{C \in Q^+} \log p_C(x) + \sum_{C \in Q^-} \log (1 - p_C(x))$$

$$P_C(x) = a_C n_+(x) + \Delta_C \sum_{t=0}^3 2^t \text{popcnt}(B_x \wedge M_{Ct})$$

56 B

per item: shared binary code

~216 B

per predicate program

0.337

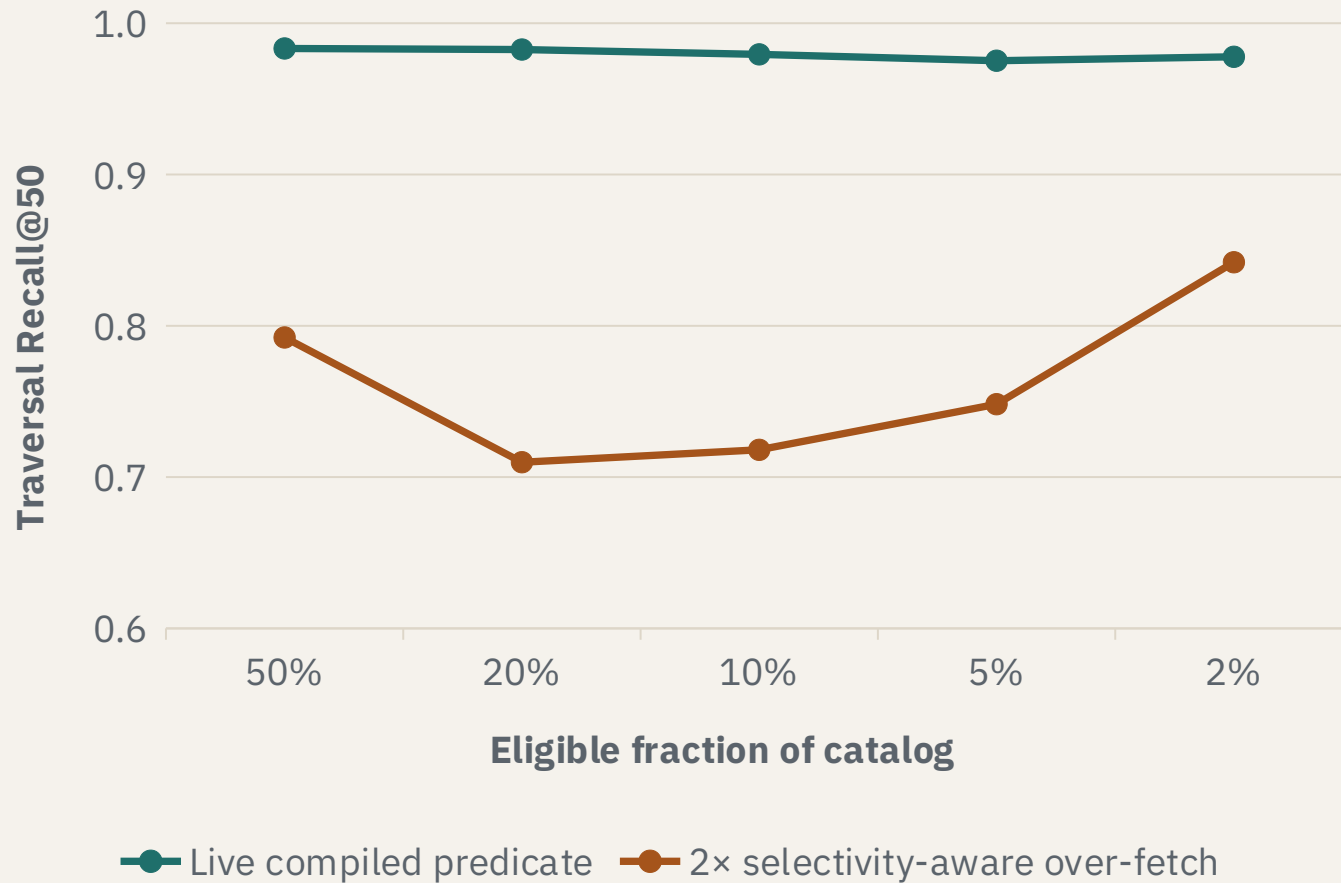
recall at 20% retention · dense 0.207
· PQ64 0.352 · FP32 0.363

96–134 ns

per predicate invocation inside HNSW

OUR RESULT IN CONTEXT

Live predicates hold recall as eligibility shrinks



AT 2% ELIGIBILITY

0.978 vs **0.842**

Recall@50, live vs 2x over-fetch

14.57 ms vs **20.14 ms**

No tested over-fetch budget got within 0.005 of live recall, in 15 of 15 conditions.

Next: measure local, not global, selectivity (anticorrelated queries).

Filtered ANN

Fast vector search under attribute filters: where RSA's predicate plugs in.

ACORN · Patel et al. 2024

Curator · Jin et al. 2026

FAVOR · Song et al. 2026

EMA · EMA authors 2026

Filtered-ANN benchmark · Iff et al. 2025

Attribute-filtering study · Li et al. 2025

Filtered ANN in vector DBs · Amanbayev et al. 2026

What it does

[Figure: ACORN, method overview figure]

Placeholder: PDF not reachable at build time. Patel et al. (2024), arXiv:2403.04871.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it shows

[Figure: ACORN, main results figure or table]

Placeholder: PDF not reachable at build time. Patel et al. (2024), arXiv:2403.04871.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it does

[Figure: Curator, method overview figure]

Placeholder: PDF not reachable at build time. Jin et al. (2026), arXiv:2601.01291.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it shows

[Figure: Curator, main results figure or table]

Placeholder: PDF not reachable at build time. Jin et al. (2026), arXiv:2601.01291.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it does

[Figure: FAVOR, method overview figure]

Placeholder: PDF not reachable at build time. Song et al. (2026), arXiv:2605.07770.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it shows

[Figure: FAVOR, main results figure or table]

Placeholder: PDF not reachable at build time. Song et al. (2026), arXiv:2605.07770.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it does

[Figure: EMA, method overview figure]

Placeholder: PDF not reachable at build time. EMA authors (2026), arXiv:2606.00734.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it shows

[Figure: EMA, main results figure or table]

Placeholder: PDF not reachable at build time. EMA authors (2026), arXiv:2606.00734.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it does

[Figure: Filtered-ANN benchmark, method overview figure]

Placeholder: PDF not reachable at build time. Iff et al. (2025), arXiv:2507.21989.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it shows

[Figure: Filtered-ANN benchmark, main results figure or table]

Placeholder: PDF not reachable at build time. Iff et al. (2025), arXiv:2507.21989.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it does

[Figure: Attribute-filtering study, method overview figure]

Placeholder: PDF not reachable at build time. Li et al. (2025), arXiv:2508.16263.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it shows

[Figure: Attribute-filtering study, main results figure or table]

Placeholder: PDF not reachable at build time. Li et al. (2025), arXiv:2508.16263.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it does

[Figure: Filtered ANN in vector DBs, method overview figure]

Placeholder: PDF not reachable at build time. Amanbayev et al. (2026), arXiv:2602.11443.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it shows

[Figure: Filtered ANN in vector DBs, main results figure or table]

Placeholder: PDF not reachable at build time. Amanbayev et al. (2026), arXiv:2602.11443.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

Filtered search is fast now; the predicate is still assumed

[Figure: ACORN, predicate-subgraph traversal / neighbor expansion, or recall vs. selectivity]

Placeholder: PDF not reachable at build time. Patel et al. (2024), arXiv:2403.04871.

OUR LIVE TRAVERSAL RULE (RSA INSIDE HNSW)

navigate by $s_{\text{dense}}(q, x)$, admit x if $S_{\text{semantic}}(x) \geq \tau$

ACORN makes HNSW predicate-agnostic by traversing the predicate subgraph.

FAVOR: 1.3–5× higher QPS at Recall@10 = 95%.

EMA: 1.68–12.25× speedup over general-filtering ANN methods.

Our read: all three take the predicate as given. A compiled RSA program can be that predicate.

Negation and logic in retrieval

Exclusion, negation and instructions: the bar for RSA's $\log(1 - p)$ algebra.

Reproducing NevIR · van den Elsen et al. 2025

E-SENS · Kim et al. 2026

EXCISE · Ali et al. 2026

Negation taxonomy · Petcu et al. 2025

Promptriever · Weller et al. 2024

mFollowIR (ECIR '25) · Weller et al. 2025

What it does

[Figure: Reproducing NevIR, method overview figure]

Placeholder: PDF not reachable at build time. van den Elsen et al. (2025), arXiv:2502.13506.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it shows

[Figure: Reproducing NevIR, main results figure or table]

Placeholder: PDF not reachable at build time. van den Elsen et al. (2025), arXiv:2502.13506.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it does

[Figure: E-SENS, method overview figure]

Placeholder: PDF not reachable at build time. Kim et al. (2026), arXiv:2608.30130.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it shows

[Figure: E-SENS, main results figure or table]

Placeholder: PDF not reachable at build time. Kim et al. (2026), arXiv:2608.30130.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it does

[Figure: EXCISE, method overview figure]

Placeholder: PDF not reachable at build time. Ali et al. (2026), arXiv:2608.05497.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it shows

[Figure: EXCISE, main results figure or table]

Placeholder: PDF not reachable at build time. Ali et al. (2026), arXiv:2608.05497.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it does

[Figure: Negation taxonomy, method overview figure]

Placeholder: PDF not reachable at build time. Petcu et al. (2025), arXiv:2507.22337.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it shows

[Figure: Negation taxonomy, main results figure or table]

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

Placeholder: PDF not reachable at build time. Petcu et al. (2025), arXiv:2507.22337.

What it does

[Figure: Promptriever, method overview figure]

Placeholder: PDF not reachable at build time. Weller et al. (2024), arXiv:2409.11136.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it shows

[Figure: Promptriever, main results figure or table]

Placeholder: PDF not reachable at build time. Weller et al. (2024), arXiv:2409.11136.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it does

[Figure: mFollowIR (ECIR '25), method overview figure]

Placeholder: PDF not reachable at build time. Weller et al. (2025), arXiv:2501.19264.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

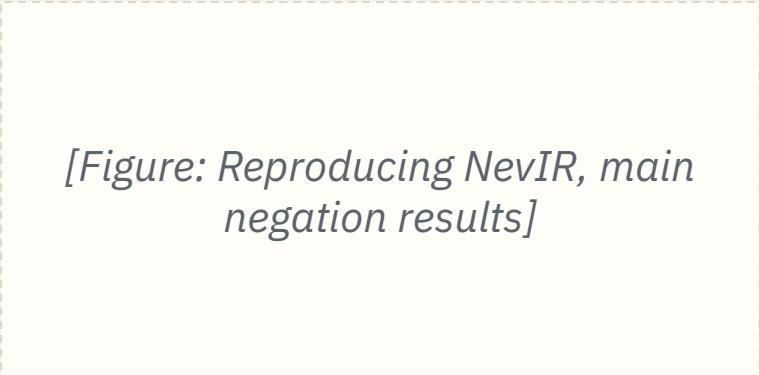
What it shows

[Figure: mFollowIR (ECIR '25), main results figure or table]

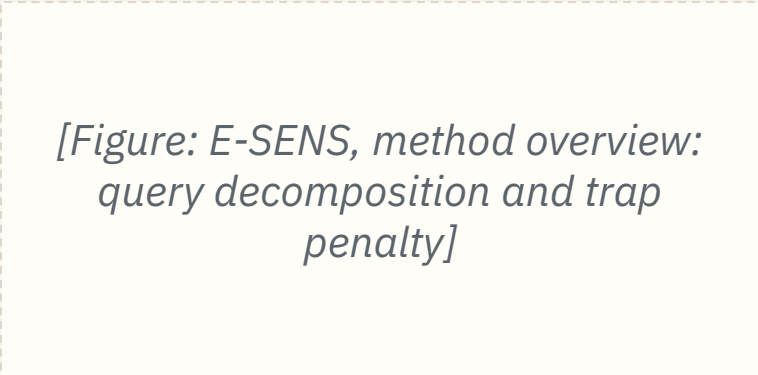
Placeholder: PDF not reachable at build time. Weller et al. (2025), arXiv:2501.19264.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

Negation is still broken; E-SENS is the bar to beat



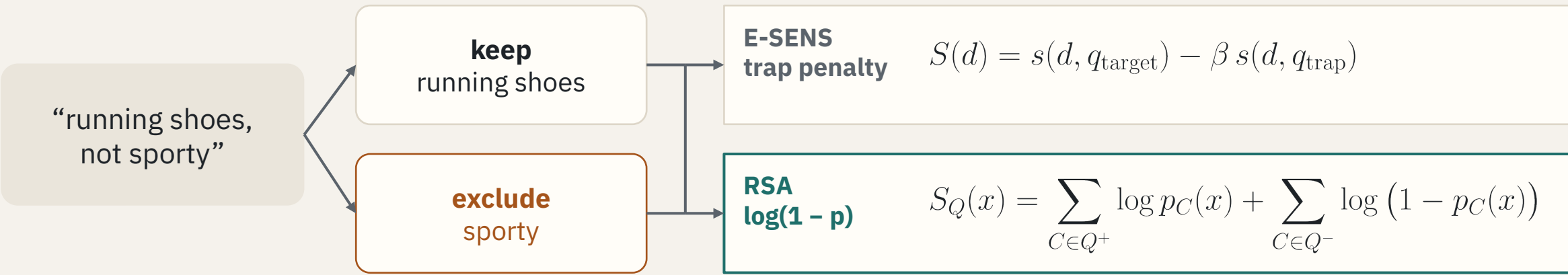
Placeholder: PDF not reachable at build time. van den Elsen et al. (2025), arXiv:2502.13506.



Placeholder: PDF not reachable at build time. Kim et al. (2026), arXiv:2608.30130.

Most retrievers still rank negated pairs at or below random.

Our bar is now **beating E-SENS**, not zero-shot directions.



Sparse autoencoders and interpretable embeddings

Concept vocabularies from embeddings: RSA predicates and map axes.

SAE embeddings toolkit · Jiang et al. 2025

Decoding dense embeddings (CL-SR) · Park et al.
2025

SAE for SPLADE · Zong et al. 2026

Xetrieval · Cai et al. 2026

What it does

[Figure: SAE embeddings toolkit, method overview figure]

Placeholder: PDF not reachable at build time. Jiang et al. (2025), arXiv:2512.10092.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it shows

[Figure: SAE embeddings toolkit, main results figure or table]

Placeholder: PDF not reachable at build time. Jiang et al. (2025), arXiv:2512.10092.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it does

[Figure: Decoding dense embeddings (CL-SR), method overview figure]

Placeholder: PDF not reachable at build time. Park et al. (2025), arXiv:2506.00041.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it shows

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

[Figure: Decoding dense embeddings (CL-SR), main results figure or table]

Placeholder: PDF not reachable at build time. Park et al. (2025), arXiv:2506.00041.

What it does

[Figure: SAE for SPLADE, method overview figure]

Placeholder: PDF not reachable at build time. Zong et al. (2026), arXiv:2604.21511.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it shows

[Figure: SAE for SPLADE, main results figure or table]

Placeholder: PDF not reachable at build time. Zong et al. (2026), arXiv:2604.21511.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it does

[Figure: Xetrieval, method overview figure]

Placeholder: PDF not reachable at build time. Cai et al. (2026), arXiv:2605.29507.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it shows

[Figure: Xetrieveal, main results figure or table]

Placeholder: PDF not reachable at build time. Cai et al. (2026), arXiv:2605.29507.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

SAEs could hand RSA a concept vocabulary for free

[Figure: SAE embeddings toolkit, Figure 1: converting documents into interpretable SAE embeddings]

Placeholder: PDF not reachable at build time. Jiang et al. (2025), arXiv:2512.10092.

SPARSE AUTOENCODER

$$z = \text{ReLU}(W_e x + b_e), \quad \mathcal{L} = \|x - W_d z\|_2^2 + \lambda \|z\|_1$$

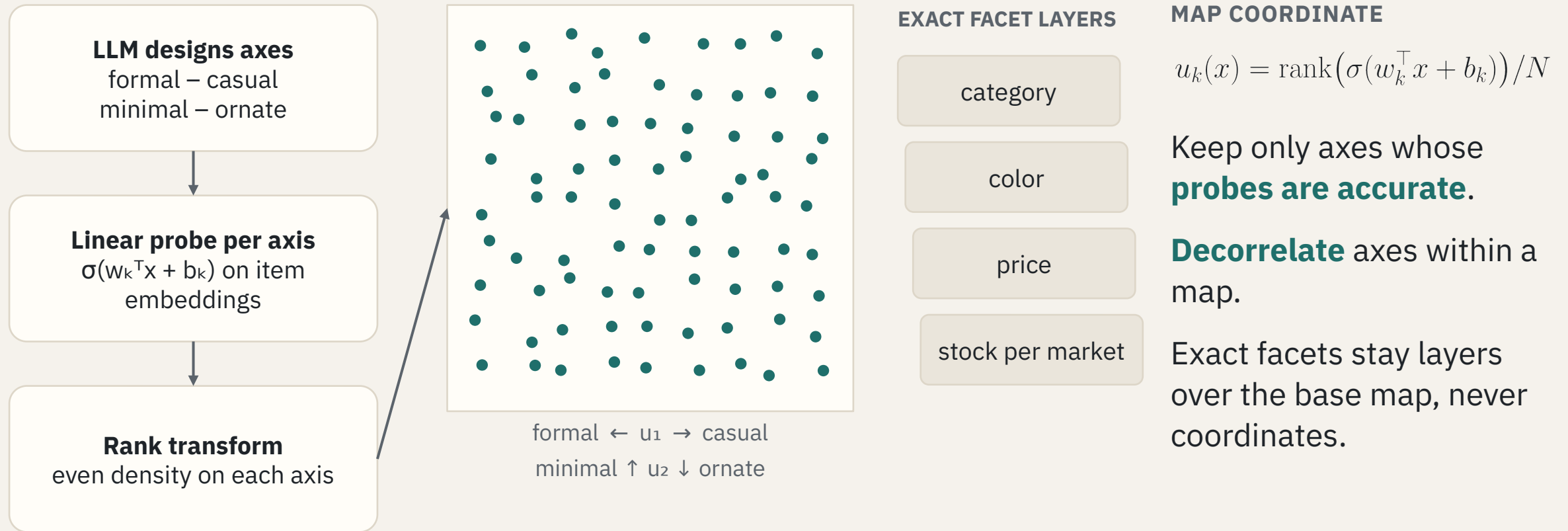
SAE features give RSA a **teacher-free** concept vocabulary.

Jiang et al. use SAE embeddings for data diffing, correlations, clustering and retrieval.

Park et al. (CL-SR) turn SAE latents into concept-level sparse retrieval.

Risks we must test: feature absorption, domain shift, extra compute.

An LLM designs the axes; probes place the items



Composed image retrieval

Edit-by-text retrieval in fashion: the prior art for the intent canvas.

Fashion IQ · Wu et al. 2019

Composed multi-modal retrieval survey · Zhang et al. 2025

FACap · Gardères et al. 2025

FashionMV · Yuan et al. 2026

FIRE-CIR (CVPR '26) · Gardères et al. 2026

Zero-shot CIR · Kakarla et al. 2025

What it does

[Figure: Fashion IQ, method overview figure]

Placeholder: PDF not reachable at build time. Wu et al. (2019), arXiv:1905.12794.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it shows

[Figure: Fashion IQ, main results figure or table]

Placeholder: PDF not reachable at build time. Wu et al. (2019), arXiv:1905.12794.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it does

[Figure: Composed multi-modal retrieval survey, method overview figure]

Placeholder: PDF not reachable at build time. Zhang et al. (2025), arXiv:2503.01334.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it shows

[Figure: Composed multi-modal retrieval survey, main results figure or table]

Placeholder: PDF not reachable at build time. Zhang et al. (2025), arXiv:2503.01334.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it does

[Figure: FACap, method overview figure]

Placeholder: PDF not reachable at build time. Gardères et al. (2025), arXiv:2507.07135.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it shows

[Figure: FACap, main results figure or table]

Placeholder: PDF not reachable at build time. Gardères et al. (2025), arXiv:2507.07135.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it does

[Figure: FashionMV, method overview figure]

Placeholder: PDF not reachable at build time. Yuan et al. (2026), arXiv:2604.10297.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it shows

[Figure: FashionMV, main results figure or table]

Placeholder: PDF not reachable at build time. Yuan et al. (2026), arXiv:2604.10297.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it does

[Figure: FIRE-CIR (CVPR '26), method overview figure]

Placeholder: PDF not reachable at build time. Gardères et al. (2026), arXiv:2604.09114.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it shows

[Figure: FIRE-CIR (CVPR '26), main results figure or table]

Placeholder: PDF not reachable at build time. Gardères et al. (2026), arXiv:2604.09114.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it does

[Figure: Zero-shot CIR, method overview figure]

Placeholder: PDF not reachable at build time. Kakarla et al. (2025), arXiv:2506.06602.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it shows

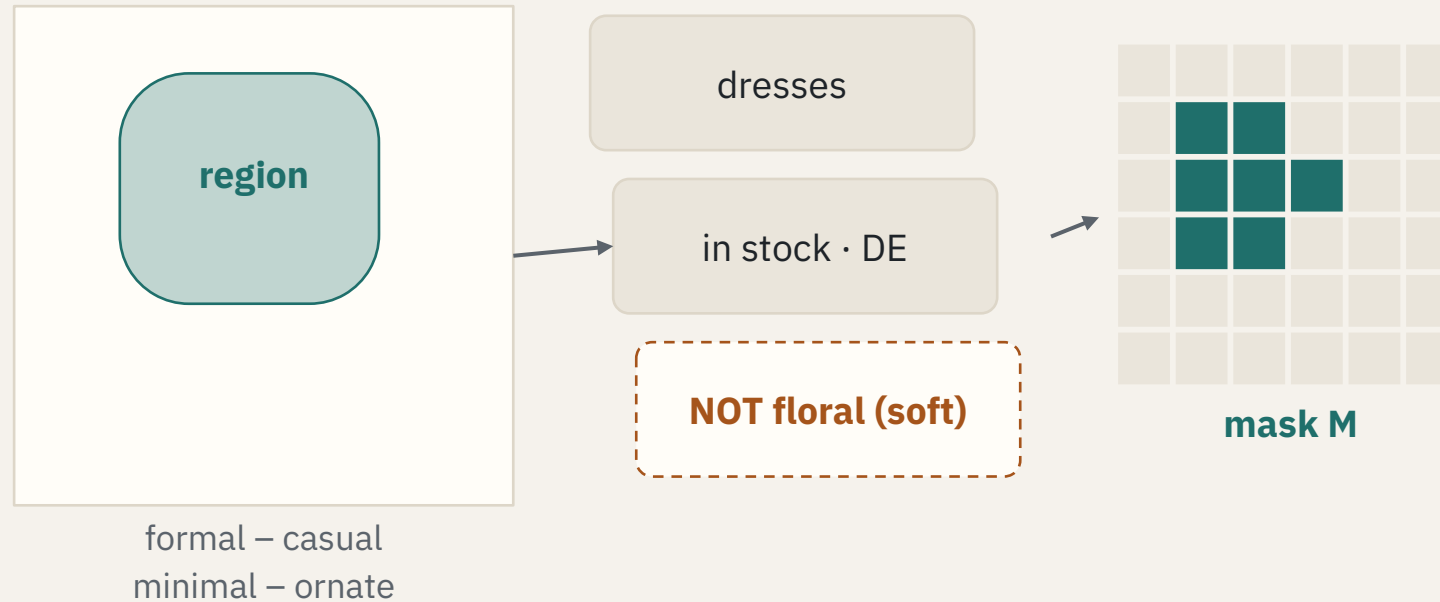
[Figure: Zero-shot CIR, main results figure or table]

Placeholder: PDF not reachable at build time. Kakarla et al. (2025), arXiv:2506.06602.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

INTENT CANVAS

Each clarification becomes layer algebra



[Figure: Fashion IQ, dataset examples: reference image, relative caption, target image]

Placeholder: PDF not reachable at build time. Wu et al. (2019), arXiv:1905.12794.

Unmentioned attributes **stay fixed** across turns.

Fashion IQ collects relative-caption feedback for dresses, shirts, tops & tees.

CANVAS MASK AS LAYER ALGEBRA

$$M = R_{\text{base}} \wedge L_{\text{dress}} \wedge L_{\text{stock,DE}} \wedge \neg L_{\text{floral}}$$

Generative retrieval in e-commerce

Semantic IDs and generated candidates: maps as codes, masks as pruning.

CQ-SID (Tmall) · Tmall authors 2026

OneSearch (Kuaishou) · Kuaishou authors 2025

OneSearch-V2 (Kuaishou) · Chen et al. 2026

VARG (Tmall) · Chu et al. 2026

FORGE · Fu et al. 2025

What it does

[Figure: CQ-SID (Tmall), method overview figure]

Placeholder: PDF not reachable at build time. Tmall authors (2026), arXiv:2605.14434.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it shows

[Figure: CQ-SID (Tmall), main results figure or table]

Placeholder: PDF not reachable at build time. Tmall authors (2026), arXiv:2605.14434.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it does

[Figure: OneSearch (Kuaishou), method overview figure]

Placeholder: PDF not reachable at build time. Kuaishou authors (2025), arXiv:2509.03236.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it shows

[Figure: OneSearch (Kuaishou), main results figure or table]

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

Placeholder: PDF not reachable at build time. Kuaishou authors (2025), arXiv:2509.03236.

What it does

[Figure: OneSearch-V2 (Kuaishou), method overview figure]

Placeholder: PDF not reachable at build time. Chen et al. (2026), arXiv:2603.24422.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it shows

[Figure: OneSearch-V2 (Kuaishou), main results figure or table]

Placeholder: PDF not reachable at build time. Chen et al. (2026), arXiv:2603.24422.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it does

[Figure: VARG (Tmall), method overview figure]

Placeholder: PDF not reachable at build time. Chu et al. (2026), arXiv:2609.14493.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it shows

[Figure: VARG (Tmall), main results figure or table]

Placeholder: PDF not reachable at build time. Chu et al. (2026), arXiv:2609.14493.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it does

[Figure: FORGE, method overview figure]

Placeholder: PDF not reachable at build time. Fu et al. (2025), arXiv:2509.20904.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

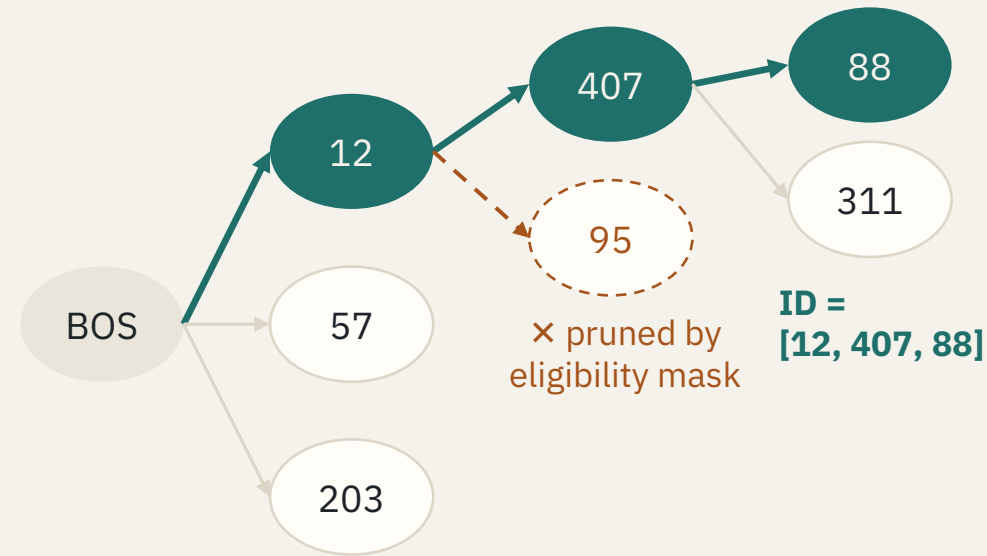
What it shows

[Figure: FORGE, main results figure or table]

Placeholder: PDF not reachable at build time. Fu et al. (2025), arXiv:2509.20904.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

Semantic IDs: generate codes, then items



Beam search over codes; highlighted path = one item.

At our 3M items the value is **different candidates**, not efficiency.

[Figure: CQ-SID (Tmall), framework overview]

Placeholder: PDF not reachable at build time. Tmall authors (2026), arXiv:2605.14434.

[Figure: OneSearch (Kuaishou), framework overview]

Placeholder: PDF not reachable at build time. Kuaishou authors (2025), arXiv:2509.03236.

+1.15%
GMV, online A/B (CQ-SID)

72.63%
of purchases via the generative channel (CQ-SID)

-75.40%
operational expenditure (OneSearch)

SEMANTIC ID BY RQ $r_0 = e(x), \quad c_\ell = \arg \min_k \|r_{\ell-1} - C_\ell[k]\|, \quad r_\ell = r_{\ell-1} - C_\ell[c_\ell], \quad \text{ID}(x) = (c_1, \dots, c_L)$

Production search, QU and conversation

How deployed systems structure queries, facets and dialogue.

GenFacet (JD.com) · Zhai et al. 2026

Case-driven multi-agent relevance · ByteDance
Global E-Commerce Search Relevance Team 2026

Conversational semantic search · Siddiqui et al.
2026

Beyond Relevance · Koushik et al. 2026

Live-catalogue conv. rec. · Kapetanovic et al. 2026

What it does

[Figure: GenFacet (JD.com), method overview figure]

Placeholder: PDF not reachable at build time. Zhai et al. (2026), arXiv:2603.19665.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it shows

[Figure: GenFacet (JD.com), main results figure or table]

Placeholder: PDF not reachable at build time. Zhai et al. (2026), arXiv:2603.19665.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it does

[Figure: Case-driven multi-agent relevance, method overview figure]

Placeholder: PDF not reachable at build time. ByteDance Global E-Commerce Search Relevance Team (2026), arXiv:2605.05991.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it shows

[Figure: Case-driven multi-agent relevance, main results figure or table]

Placeholder: PDF not reachable at build time. ByteDance Global E-Commerce Search Relevance Team (2026), arXiv:2605.05991.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it does

[Figure: Conversational semantic search, method overview figure]

Placeholder: PDF not reachable at build time. Siddiqui et al. (2026), arXiv:2601.16492.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it shows

[Figure: Conversational semantic search, main results figure or table]

Placeholder: PDF not reachable at build time. Siddiqui et al. (2026), arXiv:2601.16492.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it does

[Figure: Beyond Relevance, method overview figure]

Placeholder: PDF not reachable at build time. Koushik et al. (2026), arXiv:2609.23646.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it shows

[Figure: Beyond Relevance, main results figure or table]

Placeholder: PDF not reachable at build time. Koushik et al. (2026), arXiv:2609.23646.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it does

[Figure: Live-catalogue conv. rec., method overview figure]

Placeholder: PDF not reachable at build time. Kapetanovic et al. (2026), arXiv:2608.27006.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

What it shows

[Figure: Live-catalogue conv. rec., main results figure or table]

Placeholder: PDF not reachable at build time. Kapetanovic et al. (2026), arXiv:2608.27006.

Notes pending: figure auto-selected from the paper's captions; statements are written only after the PDF has been read (see qa_report.md).

Industry moves LLM reasoning off the query path

[Figure: Case-driven multi-agent relevance, framework overview]

Placeholder: PDF not reachable at build time. ByteDance Global E-Commerce Search Relevance Team (2026), arXiv:2605.05991.

[Figure: GenFacet (JD.com), framework overview]

Placeholder: PDF not reachable at build time. Zhai et al. (2026), arXiv:2603.19665.

GenFacet (JD.com): generated facets, +42.0% facet CTR and +2.0% UCVR online.

Case-driven relevance (ByteDance): Annotator, Optimizer and User agents learn from cases.

Our read: the same **compile-once** pattern as RSA.

FUSION

RRF is a GAM with hand-fixed shape functions

RECIPROCAL RANK FUSION

$$\text{RRF}(d) = \sum_{r \in R} \frac{1}{k + \text{rank}_r(d)}, \quad k = 60$$

FUSION AS A GAM (AN EBM GENERALIZES RRF)

$$s(d) = \sum_r f_r(\text{rank}_r(d)) + \sum_{r < r'} f_{rr'}(\text{rank}_r(d), \text{rank}_{r'}(d))$$

An EBM **learns** the shape functions that RRF fixes by hand, plus pairwise terms.

Same rank features in; our L1 already logs them.

RRF CONTRIBUTION, k = 60

rank	1 / (60 + rank)
1	0.0164
10	0.0143
30	0.0111
100	0.0063

Rank 1 counts only 2.6× rank 100: a flat, fixed curve.

Our novelty lives in two places:
compiled predicates and persistent
intent state.

Filtering speed, negation scoring and generative retrieval are crowded; we should reuse them.

Where each idea stands against the closest work

Our idea	Closest recent work	Our edge	Gap to close
Compiled predicates	SAE concept features	216 B program on a 56 B code	Beat SAE-derived predicates
Predicates in HNSW	ACORN, FAVOR, EMA	Learned predicate, 96–134 ns	Head-to-head vs ACORN
Negation algebra	E-SENS	Calibrated, composable $\log(1 - p)$	Beat E-SENS on its benchmark
Semantic maps	SAE clustering	LLM-designed, readable axes	Probe accuracy per axis
Intent canvas	Fashion IQ, FACap	Persistent, editable state	Multi-turn edit benchmark
MPDM as a retriever	CQ-SID, OneSearch	Map cells as readable codes	New candidates at 3M items

Stronger rivals to beat next

RSA

ACORN / FAVOR baseline

E-SENS on negation

SAE-derived predicates

Anticorrelated-query slice

Semantic maps

Parametric UMAP and SOM
baselines

Probe accuracy per axis

Axis decorrelation

Intent canvas

Multi-turn Fashion IQ edits

Simulated shoppers

Turns-to-target vs query
rewriting

One-market pilot

READING LIST

Papers in this deck (1/3)

OUR WORK

RSA (our paper) local PDF

FILTERED ANN

ACORN [arXiv:2403.04871](https://arxiv.org/abs/2403.04871)

Curator [arXiv:2601.01291](https://arxiv.org/abs/2601.01291)

FAVOR [arXiv:2605.07770](https://arxiv.org/abs/2605.07770)

EMA [arXiv:2606.00734](https://arxiv.org/abs/2606.00734)

Filtered-ANN benchmark [arXiv:2507.21989](https://arxiv.org/abs/2507.21989)

Attribute-filtering study [arXiv:2508.16263](https://arxiv.org/abs/2508.16263)

Filtered ANN in vector DBs [arXiv:2602.11443](https://arxiv.org/abs/2602.11443)

NEGATION AND LOGIC IN RETRIEVAL

Reproducing NevIR [arXiv:2502.13506](https://arxiv.org/abs/2502.13506)

E-SENS [arXiv:2608.30130](https://arxiv.org/abs/2608.30130)

EXCISE [arXiv:2608.05497](https://arxiv.org/abs/2608.05497)

Negation taxonomy [arXiv:2507.22337](https://arxiv.org/abs/2507.22337)

Promptriever [arXiv:2409.11136](https://arxiv.org/abs/2409.11136)

mFollowIR (ECIR '25) [arXiv:2501.19264](https://arxiv.org/abs/2501.19264)

SPARSE AUTOENCODERS AND INTERPRETABLE EMBEDDINGS

SAE embeddings toolkit [arXiv:2512.10092](https://arxiv.org/abs/2512.10092)

Decoding dense embeddings [arXiv:2506.00041](https://arxiv.org/abs/2506.00041)

READING LIST

Papers in this deck (2/3)

SAE for SPLADE [arXiv:2604.21511](https://arxiv.org/abs/2604.21511)

Xetrieval [arXiv:2605.29507](https://arxiv.org/abs/2605.29507)

COMPOSED IMAGE RETRIEVAL

Fashion IQ [arXiv:1905.12794](https://arxiv.org/abs/1905.12794)

Composed retrieval survey [arXiv:2503.01334](https://arxiv.org/abs/2503.01334)

FACap [arXiv:2507.07135](https://arxiv.org/abs/2507.07135)

FashionMV [arXiv:2604.10297](https://arxiv.org/abs/2604.10297)

FIRE-CIR (CVPR '26) [arXiv:2604.09114](https://arxiv.org/abs/2604.09114)

Zero-shot CIR [arXiv:2506.06602](https://arxiv.org/abs/2506.06602)

GENERATIVE RETRIEVAL IN E-COMMERCE

CQ-SID (Tmall) [arXiv:2605.14434](https://arxiv.org/abs/2605.14434)

OneSearch (Kuaishou) [arXiv:2509.03236](https://arxiv.org/abs/2509.03236)

OneSearch-V2 (Kuaishou) [arXiv:2603.24422](https://arxiv.org/abs/2603.24422)

VARG (Tmall) [arXiv:2609.14493](https://arxiv.org/abs/2609.14493)

FORGE [arXiv:2509.20904](https://arxiv.org/abs/2509.20904)

PRODUCTION SEARCH, QU AND CONVERSATION

GenFacet (JD.com) [arXiv:2603.19665](https://arxiv.org/abs/2603.19665)

Case-driven relevance [arXiv:2605.05991](https://arxiv.org/abs/2605.05991)

Conversational semantic search [arXiv:2601.16492](https://arxiv.org/abs/2601.16492)

Beyond Relevance [arXiv:2609.23646](https://arxiv.org/abs/2609.23646)

READING LIST

Papers in this deck (3/3)

Live-catalogue conv. rec.

[arXiv:2608.27006](#)

SOURCES (NO DEDICATED SLIDES)

[pith.science citation list for FAVOR / EMA](#)

[Voxel51: Composed image retrieval at CVPR 2025](#)

[LLMSearchRecommender compendium](#)

[interp_embed code \(Jiang et al.\)](#)